

Introduction to Inertial Navigation

Narve S. Kjørsvik

February 24, 2010

1 Introduction

The following development summarizes the derivations required to implement an integrated GNSS/INS navigation system by Kalman filtering. The filter is mechanized in the Earth Centered Earth Fixed (ECEF) frame, also denoted the e -frame. Some basic knowledge of the Kalman filter is assumed, although all the required equations are included.

1.1 General assumptions and notation

Superscripts denotes coordinatization frame. A matrix C_a^b denotes an orthonormal matrix used to transform from the a -frame to the b -frame. This matrix is also referred to as a Direction Cosine Matrix (DCM).

The index i is reserved for the inertial frame, e is reserved for the Earth frame, s is reserved for the (IMU) sensor frame, b is reserved for the body frame and g is reserved for the local level geodetic frame (North-East-Down).

The b -frame is used for relating the navigation system to other on-board sensors, e.g. a camera, lidar systems etc. In cases where such sensors are not considered, the s -frame may often be considered identical to the b -frame.

A few common coordinated transforms are included in Appendix A.

2 Inertial Measurement Units

An Inertial Measurement Unit (IMU) typically consists of accelerometer and gyro triads, properly aligned and with orthogonal sensitivity axes.

Output from the accelerometers are usually velocity increments in the sensor frame (s -frame), corrected for deterministic sensor errors (e.g. temperature-dependent effects and axis non-orthogonalities). A measurement may be written as

$$\Delta \tilde{v}^s = \int_{\delta t} \tilde{f}^s dt = \int_{\delta t} f^s + \delta f^s dt, \quad (1)$$

where the tilde denotes an observed quantity (corrupted by errors), f^s denotes the specific force, δf^s denotes instrument errors, and δt denotes the IMU data interval (e.g. 1/200 s).

Output from the gyros are usually angular increments in the s -frame, corrected for deterministic sensor errors. A measurement may be written as

$$\Delta \tilde{\theta}^s = \int_{\delta t} \tilde{\omega}_{is}^s dt = \int_{\delta t} \omega_{is}^s + \delta \omega_{is}^s dt, \quad (2)$$

where, again, tilde denotes an observed quantity, ω_{is}^s denotes angular velocity of the s -frame relative the inertial frame (i -frame), coordinatized in the s -frame, and $\delta\omega_{is}^s$ denotes instrument errors.

3 Conceptual overview of Inertial Navigation

Consider now a system with known initial conditions at time t_0 , i.e. a priori position $\tilde{x}^e(t_0)$, velocity $\tilde{\dot{x}}^e(t_0)$ and orientation in the form of a direction cosine matrix $\tilde{C}_s^e(t_0)$.

The angular increments reported by the gyros may then be used to update the sensor orientation (C_s^e). The updated DCM may then be used to transform the sensed velocity increments (in the s -frame) into corresponding velocity increments in the e -frame. Updated position and velocity estimates at time $t_0 + \delta t$ may then be obtained by a simple numerical integration procedure (single integration for the velocity, double integration for the position).

Assuming correct initial conditions and absence of instrument errors, the inertial navigation system will provide continuously updated position, velocity and orientation of the moving platform. This is denoted “Free-inertial navigation”, i.e. navigation based solely on inertial sensors, without any form of external aiding.

Unfortunately, instrument errors are always present, and due to the nature of (double) integration, the inertial navigation system is highly unstable without some sort of external aiding. Free-inertial navigation is often only possible for periods of seconds to minutes, depending on the quality of the instrument. Obviously, if the INS were able to somehow compensate for instrument errors, the performance would improve tremendously.

Global Navigation Satellite Systems (GNSS) constitute ideal aiding sensors for land and air applications. An accurate position provided by the GNSS yields an accurate estimate of the position error in the solution obtained by the INS. However, how can this observed position error be used to obtain estimates of the IMU instrument errors?

The answer to this question can be found in the theory of linear dynamic systems and optimal linear estimation, as briefly reviewed in the following two sections.

4 Linear Dynamic Systems

A comprehensive treatment of linear dynamic systems in the context of optimal estimation may be found in e.g. Gelb (1974) and Brown and Hwang (1997). The starting point is a continuous system, consisting of a set of linear non-homogeneous first-order differential equations

$$\dot{x}(t) = F(t)x(t) + G(t)u(t), \quad (3)$$

where $x(t)$ is the vector of random system variables, $F(t)$ and $G(t)$ are known matrices and $u(t)$ is a vector of zero mean, Gaussian white noise processes.

The solution of Eq. (3) w.r.t. $x(t)$ is (e.g. Gelb, 1974, pp. 55–66)

$$x(t) = \Phi(t, t_0)x(t_0) + \int_{t_0}^t \Phi(t, t')G(t')u(t')dt', \quad (4)$$

where $\Phi(t, t_0)$ is called the *state transition matrix*. For a constant F , the state transition matrix is given by

$$\Phi(t, t') = e^{F(t-t')}. \quad (5)$$

Eq. (5) can be solved using e.g. the Laplace transform, or as shown below by a Taylor series expansion:

$$\Phi(t, t') = I + F(t-t') + \frac{1}{2!}(F(t-t'))^2 + \frac{1}{3!}(F(t-t'))^3 + \dots \quad (6)$$

In the following, consider a discrete (sampled) counterpart of Eq. (4), where $x_k \equiv x(t_k)$, yielding

$$x_k = \Phi(t_k, t_{k-1})x_{k-1} + w_k, \quad (7)$$

with

$$w_k = \int_{t_{k-1}}^{t_k} \Phi(t, t')G(t')u(t')dt'. \quad (8)$$

We are now seeking an expression for the covariance matrix of w_k , i.e.

$$Q_k \equiv E\{w_k w_k^T\} = \iint_{t_{k-1}}^{t_k} \Phi(t, \xi)G(\xi)E\{u(\xi)u^T(\eta)\}G^T(\eta)\Phi^T(t, \eta)d\xi d\eta. \quad (9)$$

Let W be the power spectral density (PSD) matrix of the forcing function u . Since the auto-covariance of a white noise process is a Dirac delta function with amplitude equal to the PSD of the process, Eq. (9) simplifies to a single integral, and Q_k is formally given by

$$Q_k = \int_{t_{k-1}}^{t_k} \Phi(t, \eta)G(\eta)WG^T(\eta)\Phi^T(t, \eta)d\eta. \quad (10)$$

The error covariance of the propagated state is then

$$P_k \equiv E\{x_k x_k^T\} = \Phi(t_k, t_{k-1})P_{k-1}\Phi^T(t_k, t_{k-1}) + Q_k. \quad (11)$$

For practical purposes, Eq. (10) must be evaluated using numerical integration. For small time intervals $\delta t = (t_k - t_{k-1})$, Q_k can be approximated by a truncated Taylor series, yielding

$$\begin{aligned} Q_k \approx & GWG^T \delta t + (FGWG^T + GWG^T F^T) \frac{(\delta t)^2}{2} \\ & + (F^2GWG^T + 2FGWG^T F^T + GWG^T (F^T)^2) \frac{(\delta t)^3}{6} \\ & + (F^3GWG^T + 3F^2GWG^T F^T + 3FGWG^T (F^T)^2 \\ & \quad + GWG^T (F^T)^3) \frac{(\delta t)^4}{24} + \dots \end{aligned} \quad (12)$$

Some practical considerations regarding computation of Φ and Q_k are given in Appendix D.

5 Linear State Estimation

The previous section provided a framework for the time evolution of a vector of random variables, as well as its associated covariance matrix. This section presents the equations required to improve (update) the state estimates based on external measurements that are (approximately) linearly related to some of the state variables. The equations together form the Kalman filter. A comprehensive treatment can be found in e.g. Gelb (1974).

The equations presented here constitute the so-called extended Kalman filter (EKF). In the EKF the a priori state vector (dynamically corrected with new estimates) is used for linearization of possible non-linearities in the relation between measurements and state variables as well as in the dynamic model.

Let

$$\tilde{x}_k + \delta x_k = x_k, \quad (13)$$

describe the relation between the a priori state ($\tilde{x}_k$), the true state (x_k) and the correction (δx_k).

A vector of measurements is related to the state vector by the non-linear function

$$z_k = h(x_k) + v_k, \quad (14)$$

where v_k is a vector of random observation noise.

A linear approximation is then given by

$$\begin{aligned} \delta z_k &= z_k - h(\tilde{x}_k) \\ &= h(x_k) + v_k + h(\tilde{x}_k) \\ &\approx H_k \delta x_k + v_k, \end{aligned} \quad (15)$$

where

$$H_k = \left. \frac{\partial h}{\partial x} \right|_{x=\tilde{x}_k} \quad (16)$$

is a matrix of partial derivatives (the “design matrix”), evaluated using the a priori state.

The previous error estimate was used to update the initial state at the current epoch, and hence $\delta \tilde{x}_k \equiv 0$. This means that the time update of the state vector is trivial and can be skipped, unlike the ordinary Kalman filter.

Optimal (minimum variance) estimates of the state and the corresponding covariance matrix is then obtained by the Kalman filter, summarized by:

$$\hat{P}_k = \Phi_{k-1} \hat{P}_{k-1} \Phi_{k-1}^T + Q_k, \quad (17a)$$

$$\delta \hat{x}_k = K_k \delta z_k, \quad (17b)$$

$$\hat{P}_k = (I - K_k H_k) \tilde{P}_k, \text{ with } \quad (17c)$$

$$K_k = \tilde{P}_k H_k^T \left[H_k \tilde{P}_k + R_k \right]^{-1}, \quad (17d)$$

where P is the state covariance matrix, and under the assumptions

$$w_k \sim N(0, Q_k), \quad (18a)$$

$$v_k \sim N(0, R_k), \quad (18b)$$

$$E\{v_j w_k\} = 0, \quad \forall j, k, \quad (18c)$$

$$E\{v_j v_k\} = 0, \quad j \neq k, \quad (18d)$$

$$E\{w_j w_k\} = 0, \quad j \neq k. \quad (18e)$$

The last step of the algorithm is to update the state vector,

$$\hat{x}_k = \tilde{x}_k + \delta \hat{x}_k. \quad (19)$$

6 Navigation equations in the e -frame

The navigation equations constitute the solution to the set of differential equations relating the sensed accelerations to second-order time-derivative of the position.

A vector with coordinates x^e in the e -frame has coordinates in the i -frame given by

$$x^i = C_e^i x^e, \quad (20)$$

where C_e^i is the DCM transforming from the e -frame to the i -frame. The time derivative of a DCM can be shown (Rogers, 2007, Ch. 2.4.2, pp. 28–31) to be

$$\dot{C}_a^b = C_a^b \Omega_{ba}^a, \quad (21)$$

where

$$\Omega_{ba}^a \equiv [\omega_{ba}^a \times] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \quad (22)$$

is the angular rates of the a -frame w.r.t. the b -frame, coordinatized in the a -frame and written on skew-symmetric matrix form.

Differentiating Eq. (20) w.r.t. time and using Eq. (21) yields

$$\begin{aligned} \dot{x}^i &= \dot{C}_e^i x^e + C_e^i \dot{x}^e \\ &= C_e^i \Omega_{ie}^e x^e + C_e^i \dot{x}^e. \end{aligned} \quad (23)$$

A second differentiation of Eq. (23) w.r.t. time yields

$$\begin{aligned} \ddot{x}^i &= \dot{C}_e^i \Omega_{ie}^e x^e + C_e^i \dot{\Omega}_{ie}^e x^e + C_e^i \Omega_{ie}^e \dot{x}^e + \dot{C}_e^i \dot{x}^e + C_e^i \ddot{x}^e \\ &= C_e^i \Omega_{ie}^e \Omega_{ie}^e x^e + C_e^i \dot{\Omega}_{ie}^e x^e + C_e^i \Omega_{ie}^e \dot{x}^e + C_e^i \Omega_{ie}^e \dot{x}^e + C_e^i \ddot{x}^e \\ &= C_e^i (\Omega_{ie}^e \Omega_{ie}^e + \dot{\Omega}_{ie}^e) x^e + 2C_e^i \Omega_{ie}^e \dot{x}^e + C_e^i \ddot{x}^e. \end{aligned} \quad (24)$$

Solving Eq. (24) w.r.t. $\ddot{x}^e$ yields

$$\ddot{x}^e = C_i^e \ddot{x}^i - (\Omega_{ie}^e \Omega_{ie}^e + \dot{\Omega}_{ie}^e) x^e - 2\Omega_{ie}^e \dot{x}^e. \quad (25)$$

The fundamental relation between the specific force (f^i), inertial acceleration ($\ddot{x}^i$) and the gravitational acceleration (due to mass attraction only) g_{att}^i is (Rogers, 2007, Ch. 5.2.1, p. 104)

$$\ddot{x}^i = f^i + g_{att}^i. \quad (26)$$

Using Eq. (26) with Eq. (25) yields

$$\ddot{x}^e = C_i^e (f^i + g_{att}^i) - (\Omega_{ie}^e \Omega_{ie}^e + \dot{\Omega}_{ie}^e) x^e - 2\Omega_{ie}^e \dot{x}^e. \quad (27)$$

When assuming a constant rate of the Earth's rotation (i.e. $\dot{\Omega}_{ie}^e = 0$), we obtain

$$\begin{aligned}\ddot{x}^e &= C_i^e f^i + C_i^e g_{att}^i - \Omega_{ie}^e \Omega_{ie}^e x^e - 2\Omega_{ie}^e \dot{x}^e \\ &= f^e + g_{att}^e - \Omega_{ie}^e \Omega_{ie}^e x^e - 2\Omega_{ie}^e \dot{x}^e.\end{aligned}\quad (28)$$

By using $g_{att}^e = g^e + \Omega_{ie}^e \Omega_{ie}^e x^e$, where g^e denotes the gravity coordinatized in the e -frame, Eq. (28) may further be simplified to

$$\ddot{x}^e = f^e + g^e - 2\Omega_{ie}^e \dot{x}^e, \quad (29)$$

This system of three second order differential equations is finally transformed into a system of six first order differential equations using the velocities as new variables:

$$\frac{d}{dt} \dot{x}^e = f^e + g^e - 2\Omega_{ie}^e \dot{x}^e, \quad (30a)$$

$$\frac{d}{dt} x^e = \dot{x}^e, \quad (30b)$$

The specific force and gravitational acceleration constitute the forcing functions. By introducing known initial conditions, the system of differential equations may then be solved (by numerical integration) w.r.t. position (x^e) and velocity ($\dot{x}^e$). A 1st order algorithm for numerical integration is included in Appendix C.2.

In a strapdown INS the gyros are rigidly attached to the vehicle, and senses the angular velocities of the s -frame relative the i -frame. Correspondingly, the sensed accelerations are also provided in the s -frame, and must be transformed using C_s^e . This matrix must be integrated according to Eq. (21):

$$\dot{C}_s^e = C_s^e \Omega_{es}^s, \quad (31)$$

where the last term can be coupled to the sensed angular rates ω_{is}^s by

$$\omega_{es}^s = \omega_{is}^s - C_e^s \omega_{ie}^e. \quad (32)$$

Attitude integration is preferably formulated using quaternions, due to superior numerical stability. A brief overview of quaternions is provided in Appendix B, and a 2nd order algorithm for numerical integration of quaternions is provided in Appendix C.1. The gravity correction is usually obtained from the normal gravity model (cf. Appendix F).

7 Linearized error equations

In order to use the framework for linear dynamic systems and optimal estimation, expressions for the matrices in Eq. (3) must be derived.

Let now the variables be linearly perturbed according to

$$\tilde{x}^e = x^e - \delta x^e, \quad (33a)$$

$$\tilde{\dot{x}}^e = \dot{x}^e - \delta \dot{x}^e, \quad (33b)$$

$$\tilde{f}^s = f^s + b_f^s - \tilde{b}_f^s = f^s + \delta f^s, \quad (33c)$$

$$\tilde{\omega}_{is}^s = \omega_{is}^s + b_\omega^s - \tilde{b}_\omega^s = \omega_{is}^s + \delta \omega_{is}^s, \quad (33d)$$

$$\tilde{C}_s^e = (I - \Psi^e) C_s^e, \quad (33e)$$

$$\tilde{g}^e = g^e - \delta g^e, \quad (33f)$$

where $\Psi^e \equiv \psi^e \times$ and b_f^s and b_ω^s denotes true errors in the specific force and gyro rates respectively. Application of a priori bias estimates $\tilde{b}_f^s$ and $\tilde{b}_\omega^s$ then yields the residual inertial instrument errors δf^s and $\delta \omega^s$.

The starting point for the velocity error is Eq. (30a). We assume a strapdown INS, i.e. the accelerations are sensed in the s -frame and transformed to the e -frame by $f^e = C_s^e f^s$.

The velocity error equation is now derived by substituting the perturbed variables as defined in Eqs. (33a)–(33f) into Eq. (30a), and then subtracting Eq. (30a) using the true values.

The perturbed velocity equation will then read

$$\frac{d}{dt} \delta \dot{x}^e = -2\Omega_{ie}^e \delta \dot{x}^e + \Psi C_s^e f^s - C_s^e \delta f^s + \delta g^e, \quad (34)$$

where δg^e represents the errors in the gravity correction. This error is partly due to an erroneous position used in the computation, and partly due to imperfections in the model. In total, this can be represented by

$$\delta g^e = \Gamma^e \delta x^e - \Omega_{ie}^e \Omega_{ie}^e \delta x^e + \delta g_\epsilon^e, \quad (35)$$

where $\Gamma^e \equiv \partial g_{att}^e / \partial x^e$, and δg_ϵ^e represents a small gravity anomaly. Refer to Appendix E for a definition of Γ^e . Using Eq. (35) for the gravity errors and the rule $a \times b = -b \times a$, we arrive at the final expression for the velocity error:

$$\frac{d}{dt} \delta \dot{x}^e = -2\Omega_{ie}^e \delta \dot{x}^e + (\Gamma^e - \Omega_{ie}^e \Omega_{ie}^e) \delta x^e - (f^e \times) \psi^e - C_s^e \delta f^s + \delta g_\epsilon^e. \quad (36)$$

The position error equation follows trivially from differential perturbation of Eq. (30b):

$$\frac{d}{dt} \delta x^e = \delta \dot{x}^e. \quad (37)$$

The last derivation concerns the attitude errors. By differentially perturbing Eq. (31) we obtain

$$\delta \dot{C}_s^e = \delta C_s^e \Omega_{es}^s + C_s^e \delta \Omega_{es}^s, \quad (38)$$

with $\delta \Omega_{es}^s \equiv \Omega_{es}^s - \tilde{\Omega}_{es}^s$. We have by definition that

$$\delta C_s^e \equiv C_s^e - \tilde{C}_s^e = \Psi^e C_s^e. \quad (39)$$

Now equating Eq. (38) with the time-derivative of Eq. (39):

$$\begin{aligned} \dot{\Psi}^e C_s^e + \Psi^e C_s^e \Omega_{es}^s &= \delta C_s^e \Omega_{es}^s + C_s^e \delta \Omega_{es}^s \\ \dot{\Psi}^e C_s^e &= C_s^e \delta \Omega_{es}^s \\ \dot{\Psi}^e &= C_s^e \delta \Omega_{es}^s C_e^s, \text{ and on vector form} \\ \dot{\psi}^e &= C_s^e \delta \omega_{es}^s. \end{aligned} \quad (40)$$

The angular rate sensed by the strapdown gyros may be decomposed as

$$\omega_{is}^s = \omega_{ie}^s + \omega_{es}^s = C_e^s \omega_{ie}^e + \omega_{es}^s. \quad (41)$$

Solving for $\delta \omega_{es}^s$ and applying perturbations according to Eqs. (33d)–(33e) we obtain

$$\delta \omega_{es}^s = -\delta \omega_{is}^s + C_e^s \delta \omega_{ie}^e + C_e^s \Psi^e \omega_{ie}^e. \quad (42)$$

In this context the earth rate can be assumed free of errors ($\delta\omega_{ie}^e \equiv 0$). After changing order of the cross product we obtain

$$\delta\omega_{es}^s = -\delta\omega_{is}^s - C_e^s \Omega_{ie}^e \psi^e. \quad (43)$$

Eq. (43) may then be substituted into Eq. (40), yielding

$$\begin{aligned} \dot{\psi}^e &= C_s^e (-C_e^s \Omega_{ie}^e \psi^e - \delta\omega_{is}^s) \\ &= -\Omega_{ie}^e \psi^e - C_s^e \delta\omega_{is}^s, \end{aligned} \quad (44)$$

which is the final expression for the attitude error.

The velocity, position and attitude error equations may then be summarized in matrix form as

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \delta x^e \\ \delta \dot{x}^e \\ \psi^e \end{bmatrix} &= \begin{bmatrix} 0 & I & 0 \\ (\Gamma^e - \Omega_{ie}^e \Omega_{ie}^e) & -2\Omega_{ie}^e & -f^e \times \\ 0 & 0 & -\Omega_{ie}^e \end{bmatrix} \begin{bmatrix} \delta x^e \\ \delta \dot{x}^e \\ \psi^e \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 & 0 & 0 \\ 0 & -C_s^e & I \\ -C_s^e & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta\omega_{is}^s \\ \delta f^s \\ \delta g_\epsilon^e \end{bmatrix}. \end{aligned} \quad (45)$$

This result is easily recognized as being exactly on the form of Eq. (3), i.e. suitable for developing the matrices required for the error propagation (time update) in the Kalman filter.

8 Aided inertial navigation

Starting with an initial navigation state ($\tilde{x}^e(t_0)$, $\tilde{\dot{x}}^e(t_0)$ and $\tilde{C}_s^e(t_0)$), the navigation state at the next epoch ($t_0 + \delta t$) is developed using the sensed velocity and angular increments and by numerically integrating the navigation state (cf. Appendix C).

Due to imperfect a priori knowledge of the initial state, instrument biases and noise, etc., the integrated navigation state will diverge from the true navigation state. By including external information about one or more of the system states (e.g. a GNSS position fix), the errors of the navigation state can be estimated and corrected.

When an estimate of the navigation state error is available, the navigation state is corrected (analogously to Eqs. (33a)–(33f)) according to

$$\hat{x}^e = \tilde{x}^e + \delta \hat{x}^e, \quad (46a)$$

$$\hat{\dot{x}}^e = \tilde{\dot{x}}^e + \delta \hat{\dot{x}}^e, \quad (46b)$$

$$\hat{C}_s^e = (I + \hat{\Psi}^e) \tilde{C}_s^e, \quad (46c)$$

and possible bias states according to

$$\hat{b}_f^s = \tilde{b}_f^s + \delta \hat{b}_f^s, \quad (47a)$$

$$\hat{b}_\omega^s = \tilde{b}_\omega^s + \delta \hat{b}_\omega^s, \quad (47b)$$

$$\hat{g}^e = \tilde{g}^e + \delta \hat{g}^e. \quad (47c)$$

In a typical setup, it is the *differences* between the INS states and the aiding system states that enter the Kalman filter. Hence, the actual platform dynamics do not enter the system, only the *errors* of the navigation system. This is an effective means to obtain a linear system and to avoid complex filter tuning. This design is termed a *complementary filter*.

GNSS position and velocity measurements are commonly used to aid inertial navigation systems. Observations equations for GNSS aiding are derived in Appendix G.

A flowchart depicting an aided inertial navigation system is shown in Figure 1, complete with references to the required equations.

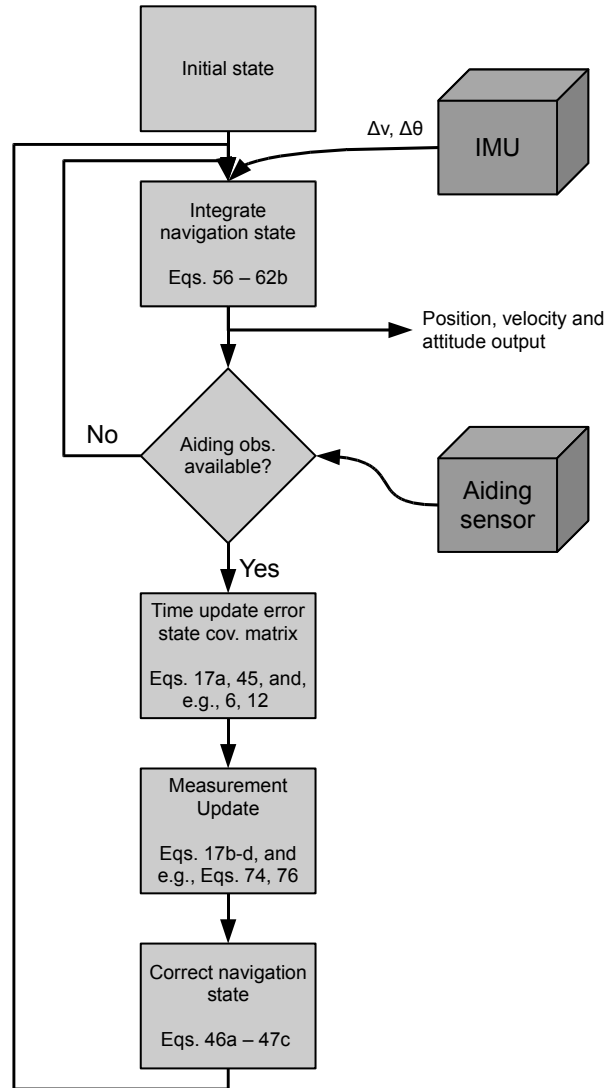


Figure 1: Aided inertial navigation.

A Common coordinate transformations

Based purely on geometrical considerations, a few common coordinate transformations will be described. A full derivation is given in e.g. (Rogers, 2007, Ch. 3).

The DCM transforming from the e -frame (Earth centered, Earth fixed) to the local level geodetic frame is given by

$$C_e^g = \begin{bmatrix} -\sin(\varphi) \cos(\lambda) & -\sin(\varphi) \sin(\lambda) & \cos(\varphi) \\ -\sin(\lambda) & \cos(\lambda) & 0 \\ -\cos(\varphi) \cos(\lambda) & -\cos(\varphi) \sin(\lambda) & -\sin(\varphi) \end{bmatrix}, \quad (48)$$

where φ and λ denotes geodetic latitude and longitude respectively.

The DCM transforming from the b -frame to the g -frame is given by

$$C_b^g = \begin{bmatrix} c(\theta) c(\alpha) & -c(\phi) s(\alpha) + s(\phi) s(\theta) c(\alpha) & s(\phi) s(\alpha) + c(\phi) s(\theta) c(\alpha) \\ c(\theta) s(\alpha) & c(\phi) c(\alpha) + s(\phi) s(\theta) s(\alpha) & -s(\phi) c(\alpha) + c(\phi) s(\theta) s(\alpha) \\ -s(\theta) & s(\phi) c(\theta) & c(\phi) c(\theta) \end{bmatrix}, \quad (49)$$

where ϕ denotes roll, θ denotes pitch, α denotes heading, and $c(\cdot)$ and $s(\cdot)$ are short-hand notations for $\cos(\cdot)$ and $\sin(\cdot)$ respectively.

In many cases, the s -frame can be considered to constitute the b -frame, in particular when no other external sensors are involved.

The DCMs may be combined (by pre-multiplication) to obtain e.g. $C_b^e = C_g^e C_b^g$.

B Quaternions

Quaternions may in some senses be thought of as a generalization of complex numbers. Quaternions are four-dimensional vectors, capable of effective and numerically stable representing a rotation in 3 dimensions. As such, quaternions constitute an alternative to direction cosine matrices for attitude representation. In the following, only *unit quaternions* are considered. A brief introduction can be found in (e.g. Jekeli, 2001, pp. 13–22).

Let c_{ij} represent the elements of the DCM C_t^s . The corresponding quaternion elements may then be constructed by

$$a = \frac{1}{2} \sqrt{1 + c_{11} + c_{22} + c_{33}}, \quad (50a)$$

$$b = \frac{1}{4a} (c_{23} - c_{32}), \quad (50b)$$

$$c = \frac{1}{4a} (c_{31} - c_{13}), \quad (50c)$$

$$d = \frac{1}{4a} (c_{12} - c_{21}), \quad (50d)$$

$$(50e)$$

and finally the full quaternion

$$q_t^s = (a, b, c, d)^T. \quad (51)$$

Correspondingly, given a quaternion as defined in Eq. (51), the DCM can be constructed as

$$C_t^s = \begin{bmatrix} a^2 + b^2 - c^2 - d^2 & 2(bc + da) & 2(bd - ca) \\ 2(bc - da) & a^2 + c^2 - b^2 - d^2 & 2(cd + ab) \\ 2(bd + ca) & 2(cd - ab) & a^2 + d^2 - b^2 - c^2 \end{bmatrix}. \quad (52)$$

The inverse (conjugate) quaternion is defined by

$$q_s^t = q_t^{s*} \equiv (a, -b, -c, -d)^T. \quad (53)$$

The quaternion differential equations (analog to Eq. (21) for DCMs) read

$$\dot{q} = \frac{1}{2} \begin{bmatrix} 0 & \omega_1 & \omega_2 & \omega_3 \\ -\omega_1 & 0 & \omega_3 & -\omega_2 \\ -\omega_2 & -\omega_3 & 0 & \omega_1 \\ -\omega_3 & \omega_2 & -\omega_1 & 0 \end{bmatrix} q. \quad (54)$$

Due to accumulated numerical errors, the quaternion must occasionally be normalized in order to satisfy unit length properties. The normalization can easily be done by

$$q \leftarrow \frac{q}{\|q\|}, \quad (55)$$

where $\leftarrow$ denotes “is replaced by”.

C Numerical Integration

Numerical integration is required for updating the navigation equations as new IMU observations arrive. The quaternion integration algorithm shown below is of 2nd order (i.e. errors are of 3rd order). The position and velocity integration algorithms described below are 1st order (trapezoid integration), i.e. (algorithm) errors are of second order. These algorithms may suffice for high data rates and moderate dynamics.

C.1 Numerical integration of quaternions

The following equations are taken from (Jekeli, 2001, pp. 115–118). Define

$$\delta\beta_k = \Delta\theta_{is}^s(t_k) - C_e^s(t_{k-1})\omega_{ie}^e\delta t, \quad (56)$$

where k is the time index, $\Delta\theta_{is}^s(t_k)$ is the gyro reading, and δt is the gyro measurement interval. $C_e^s(t_{k-1})$ can be found from the quaternion at t_{k-1} using Eq. (52).

The updated quaternion may then be computed by

$$q(t_k) = \left[\cos\left(\frac{1}{2}|\delta\beta_k|\right) I + \frac{1}{|\delta\beta_k|} \sin\left(\frac{1}{2}|\delta\beta_k|\right) B_k \right] q(t_{k-1}), \quad (57)$$

with

$$B_k = \begin{bmatrix} 0 & (\delta\beta_1)_k & (\delta\beta_2)_k & (\delta\beta_3)_k \\ -(\delta\beta_1)_k & 0 & (\delta\beta_3)_k & -(\delta\beta_2)_k \\ -(\delta\beta_2)_k & -(\delta\beta_3)_k & 0 & (\delta\beta_1)_k \\ -(\delta\beta_3)_k & (\delta\beta_2)_k & -(\delta\beta_1)_k & 0 \end{bmatrix}. \quad (58)$$

C.2 Numerical integration of position and velocity

The velocity increments in the s -frame must be transformed to the e -frame. The *average* DCM over the update interval (δt) is used, i.e.

$$\Delta v^e = \bar{C}_s^e \Delta v^s - (2\Omega_{ie}^e v^e - C_g^e g^g) \delta t, \quad (59)$$

where

$$\bar{C}_s^e = C_s^e(t_{k+1}) \left[I - \frac{1}{2}(\rho^s \times) \right], \quad (60)$$

and

$$\rho^s = \Delta \theta_{is}^s - C_e^s \omega_{ie}^e \delta t. \quad (61)$$

In Eq. (59) the velocity from the previous update can be used as an approximation in the Coriolis correction term.

Finally, the position and velocity can be updated according to

$$x^e(t_{k+1}) = x^e(t_k) + \dot{x}^e(t_k) \delta t + \Delta v^e \frac{\delta t}{2} \quad (62a)$$

$$\dot{x}^e(t_{k+1}) = \dot{x}^e(t_k) + \Delta v^e. \quad (62b)$$

It can be seen that the position is updated using the *average* velocity over the interval (velocity at the midpoint).

D Practical considerations for numerical evaluation of Φ and Q

A time interval ($t_k - t_0$) may be further subdivided in order to (better) fulfill the requirement of a constant F , and may be computed as

$$\Phi(t_k, t_0) = \Phi(t_k, t_{k-1}) \Phi(t_{k-1}, t_{k-2}) \cdots \Phi(t_1, t_0). \quad (63)$$

The same reasoning is used when approximating the process noise covariance matrix, but using a sum of integrals. Within each integral i (over an interval $\delta t = t_k - t_{k-1}$), the matrices F_i and G_i are both considered constant (cf. Eq. (12)). The result is given as

$$\begin{aligned} Q_k &= \int_{t_0}^{t_k} \Phi(t') G(t') W G^T(t') \Phi(t') dt' \\ &\approx \sum_i \left(G_i W G_i^T \delta t + (F_i G_i W G_i^T + G_i W G_i^T F_i^T) \frac{\delta t^2}{2} + \dots \right) \end{aligned} \quad (64)$$

For moderate dynamics, F_i and G_i may usually be considered constant over fractions of a seconds (e.g. 0.1 – 0.2 s). In addition, the transition matrix may occasionally be approximated by retaining only a few lower order terms in the Taylor series.

E Gravitational gradient tensor

The gravitational gradient tensor is given by (Rogers, 2007, p. 292) as

$$\frac{\partial g_{att}^e}{\partial x} = \omega_s^2 [3ee^T - I], \quad (65)$$

where e is the ECEF unit vector ($x^e/\|x^e\|$), and ω_s is the Schuler frequency, given for a non-accelerating platform by

$$\omega_s = \sqrt{\frac{\|g\|}{R_e}}, \quad (66)$$

where R_e is the radius of the earth.

F Normal gravity

The normal gravity model may often be sufficiently accurate. Following (Hofmann-Wellenhof and Moritz, 2006, p. 80), the vertical component of the normal gravity can be computed by

$$\gamma = \gamma_a(1 + f_2 \sin^2 \phi + f_4 \sin^4 \phi), \quad (67)$$

where ϕ is the latitude and

$$f_2 = -f\frac{5}{2}m + \frac{1}{2}f^2 - \frac{26}{7}fm + \frac{15}{4}m^2, \quad (68a)$$

$$f_4 = -\frac{1}{2}f^2 + \frac{5}{2}fm, \quad (68b)$$

and γ_a , f and m are constants defined for a particular ellipsoid. Correspondingly, the normal gravity at height h can be computed by (Hofmann-Wellenhof and Moritz, 2006, p. 82)

$$\gamma(h) = \gamma \left[1 - \frac{2}{a}(1 + f + m - 2f \sin^2 \phi)h + \frac{3}{a^2}h^2 \right], \quad (69)$$

where a is the semimajor axis. Due to the flattening of the earth, there will also be a small north-south component (Hofmann-Wellenhof and Moritz, 2006, p. 233)

$$\gamma_N = -\frac{f_2 + f_4}{R_e}h \sin 2\phi, \quad (70)$$

where R_e is the approximate radius of the Earth.

G GNSS position and velocity aiding

This section derives the observation equations required for use with GNSS position and velocity measurements (so-called “loose coupling”).

Positions provided by a GNSS (x_G^e) are related to the GNSS antenna, displaced from the IMU reference point by a lever-arm according to

$$x^e = x_G^e - C_s^e l^s, \quad (71)$$

where in this context the lever-arm l^s is considered approximately known and constant. The time-derivative of Eq. (71) yields the corresponding relation between velocity at the IMU and velocity at the GNSS antenna:

$$\dot{x}^e = \dot{x}_G^e - C_s^e \Omega_{es}^s l^s, \quad (72)$$

with Ω_{es}^s as defined by Eq. (32), expressed in skew-symmetric matrix form.

A GNSS position measurement vector corrupted by an error ε_p^e may be written as

$$z_p = x^e + C_s^e l^s + \varepsilon_p^e. \quad (73)$$

The linear observation equations may then be formed by subtracting the computed observations (based on a priori state variables):

$$\begin{aligned} \delta z_p &= (x^e + C_s^e l^s + \varepsilon_p^e) - (\tilde{x}^e + \tilde{C}_s^e \tilde{l}^s) \\ &\approx \delta x^e - (l^e \times) \psi^e + C_s^e \delta l^s + \varepsilon_p^e. \end{aligned} \quad (74)$$

Correspondingly, a GNSS velocity measurement vector corrupted by an error ε_v^e may be written as

$$z_v = \dot{x}^e + C_s^e \Omega_{es}^s l^s + \varepsilon_v^e. \quad (75)$$

The linear observation equations may then be formed by subtracting the computed observations (based on a priori state variables):

$$\begin{aligned} \delta z_v &= (\dot{x}^e + C_s^e \Omega_{es}^s l^s + \varepsilon_v^e) - (\tilde{\dot{x}}^e + \tilde{C}_s^e \tilde{\Omega}_{es}^s \tilde{l}^s) \\ &\approx \delta \dot{x}^e + [l^e \times \omega_{is}^e] \times \psi^e + C_s^e \Omega_{es}^s \delta l^s + C_s^e (l^s \times) \delta \omega_{is}^s + \varepsilon_v^e, \end{aligned} \quad (76)$$

where $\delta \omega_{is}^s$ denotes the residual gyro biases, as earlier defined in Eq. (33d). In Eq. (76) it has been assumed that $\|\omega_{is}^s\| \gg \|\omega_{ie}^e\|$.

Errors in the GNSS measurements (ε_p^e and ε_v^e) may be treated as white noise only, or as a linear combination of colored and white noise. Colored noise can be handled by augmenting the error state with additional variables.

Observation equation for range and range-rate observations can be derived from Eqs. (74) and (76). A range observation can be written

$$R = \rho + \eta, \quad (77)$$

where $\rho \equiv \|x^e - r^e\|$ and η denotes other range biases and errors. r^e denotes the ranging source (satellite) position. It can be shown that

$$h \equiv \frac{\partial R}{\partial \delta x^e} = \frac{(x^e - r^e)^T}{\|x^e - r^e\|}. \quad (78)$$

This result may then be combined with Eq. (74) into a complete observation equation for range aiding.

In the general case, the current epoch of the navigation state (as developed by integrating the IMU data) does not correspond to the epoch of the GNSS observations. Prior to forming the observation equations above, a navigation state for the correct epoch must be obtained by interpolation/extrapolation of the current navigation states.

References

- Brown, R. G. and Hwang, P. Y. C. (1997). *Introduction to random signals and applied Kalman filtering*. John Wiley & Sons, 3rd edition. ISBN 0471128392.
- Gelb, A. (editor) (1974). *Applied Optimal Estimation*. The M.I.T. Press. ISBN 0262570483.
- Hofmann-Wellenhof, B. and Moritz, H. (2006). *Physical geodesy*. Springer Wien New York, 2nd edition. ISBN 3211335447.
- Jekeli, C. (2001). *Inertial navigation systems with geodetic applications*. Walter de Gruyter. ISBN 3110159031.
- Rogers, R. M. (2007). *Applied Mathematics in Integrated Navigation Systems*. AIAA Education Series. American Institute of Aeronautics and Astronautics, Inc., 3rd edition. ISBN 1563479273.